RISHI RAJ

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EDUCATION

Trident Academy of Technology (TAT), Bhubaneswar — **2023 – 2027**
B.Tech in Computer Science and Engineering (Data Science) — CGPA: **7.5/10**
Coursework: Statistics · Predictive Analysis · Database Management · Data Mining & Warehousing · Machine Learning · AI

TECHNICAL SKILLS

Languages & Databases:	**SQL** (PostgreSQL – JOINs, Subqueries, Aggregations), **Python**, Relational Data Modeling, Supabase
Data Analysis & Statistics:	Data Cleaning, **EDA**, Descriptive & Inferential Statistics, Regression, RFM & Customer Segmentation, Time Series
BI & Visualization:	**Power BI**, **Microsoft Excel** (Formulas, Charts), KPI Dashboards, Matplotlib, Seaborn, Folium, Streamlit
Python & ML Libraries:	**Pandas**, **NumPy**, Scikit-learn, XGBoost, LightGBM, Statsmodels (SARIMA), DistilBERT (NLP), Selenium
Tools & Deployment:	Jupyter Notebook, Git, GitHub, GitHub Actions, Flask (REST API), Render, OSMnx, Google OR-Tools

EXPERIENCE

Data Analyst Intern | *Mind Web Ventures (Remote)* — **Jun 2026 – Aug 2026**

Objective: *Convert spatial, temporal and business data into insights for locational intelligence, predictive analysis and ESG risk reporting.*

- Performed **exploratory data analysis (EDA)** on **location-based and time-series data** to uncover patterns powering **locational intelligence**.
- Analyzed **business and industrial datasets** to identify **trends, anomalies and risk patterns** that fed into **predictive analysis** workflows.
- Prepared **analytical reports** on **risk and ESG metrics**, translating raw data into clear, concise, **business-ready insights** for decision makers.

Database Intern | *Zivanta Analytics (Remote)* — **Jun 2025 – Jul 2025**

Objective: *Support relational database operations for structured data storage, retrieval and reporting, working primarily with PostgreSQL.*

- Wrote **PostgreSQL queries** with **JOINs, subqueries and aggregations** to retrieve, transform and summarize data for **business reporting**.
- Designed and maintained **relational tables** with **primary/foreign keys and constraints** to store structured business data reliably and at scale.
- Performed **data validation** and integrity checks to catch **inconsistencies and integrity issues** before data reached downstream reports.

PROJECTS

E-Commerce Customer Retention & CLV Analytics | *PostgreSQL, SQL, Power BI* — **Oct 2024 – Dec 2024**

- Developed an end-to-end **customer retention analytics pipeline** in **PostgreSQL** with **RFM segmentation** and **CLV reporting** in **Power BI**.
- Designed an **8+ table PostgreSQL schema** for ∼**100K Olist orders**; validated **row counts and referential integrity** after every data load.
- Ran **RFM analysis** to segment customers into **Loyal, At-Risk and Lost/Churned** cohorts, enabling focused, data-backed **retention plans**.
- Built a 2-page interactive **Power BI dashboard** tracking retention **KPIs**; surfaced a critical ∼**3% repeat-purchase rate** as the key insight.

E-Commerce Review Intelligence Dashboard | *Python, Selenium, DistilBERT, Streamlit* — **Feb 2025 – Apr 2025**

- Built an **NLP-powered review analytics dashboard** that flags **rating–sentiment mismatches** to expose hidden **customer dissatisfaction**.
- Scraped **977 Flipkart smartwatch reviews** across **5 brands** using **Selenium** and stored the cleaned review data in **PostgreSQL (Supabase)**.
- Applied a pretrained **DistilBERT** sentiment model and found a ∼**2x higher mismatch rate** for one brand compared with the other four brands.
- Delivered an interactive **Streamlit dashboard** with **brand filters and side-by-side comparisons**, kept live via automated **GitHub Actions**.

Hospital Readmission Prediction | *Python, Pandas, Scikit-learn, Flask* — **Sep 2025 – Nov 2025**

- Cleaned and analyzed ∼**101K patient records** (UCI Diabetes dataset, ∼**50 features**) to model **hospital readmission risk** from clinical history.
- Engineered **diagnosis and prior-visit features**; identified **number_inpatient** as the **top risk driver** through feature-importance analysis.
- Achieved ∼**0.684 ROC-AUC**; deployed the model through a **Flask API** and documented **data ethics** considerations for sensitive patient data.

Credit Risk & Loan Default Prediction | *Python, Scikit-learn, XGBoost, Flask* — **Dec 2025 – Feb 2026**

- Cleaned ∼**150K borrower records**, treating **missing values, outliers and sentinel codes** to build a reliable, analysis-ready credit dataset.
- Engineered features and benchmarked **Logistic Regression, Random Forest and XGBoost** classifiers to predict **loan default** risk accurately.
- Reached ∼**0.86 ROC-AUC** and ∼**77% recall** on defaulters; deployed the end-to-end scoring model as a live **Flask REST API** on **Render**.

Retail Sales Forecasting (Time Series) | *Python, XGBoost, SARIMA* — **Mar 2026 – May 2026**

- Built a **sales forecasting system** comparing **SARIMA** against **ML models** on ∼**1.02M daily sales records** from **1,115 Rossmann stores**.
- Performed **EDA** on 2013–2015 sales with **store-level attributes** to uncover **seasonality, demand trends** and store-wise performance gaps.
- Engineered **lag and rolling-window features** and applied leakage-safe **TimeSeriesSplit** cross-validation for unbiased model evaluation.
- Achieved ∼**9.49% MAPE** with **XGBoost**, a ∼**34% MAE improvement** over the SARIMA baseline; deployed as a live forecasting web app.

Multi-Modal Transit & Logistics Route Optimization | *Python, OSMnx, OR-Tools* — **Sep 2026 – Oct 2026**

- Extracted **Indore's road network (∼30K nodes)** via **OSMnx** and solved a **Capacitated VRP** for a **4-vehicle delivery fleet** on real-world data.
- Benchmarked **OR-Tools** against Nearest-Neighbor (**21.8% shorter routes**) and **A*** against Dijkstra (∼**3.25x faster** shortest-path search).
- Visualized optimized routes on **Folium maps** and deployed a live **Streamlit dashboard** with **per-vehicle filters** for route-level fleet analysis.

CERTIFICATIONS & ACHIEVEMENTS

- **Google (Coursera)** – Foundations of Data Science, The Power of Statistics, Regression Analysis, Nuts & Bolts of ML — **Mar 2026**
- **Excel Basics for Beginners** (Coursera, Knowledge Accelerators) – Workbooks, Formatting, Charts and Core Skills — **May 2026**
- **HackerRank** – **5-star** in **SQL** and **4-star** in **Python**; SQL and Python Certifications — View Profile

POSITION OF RESPONSIBILITY

Core Member | *Speech Weavers Club, Trident Academy of Technology* — **Jul 2024 – May 2025**

- Managed the club's **social media** on **LinkedIn, Instagram and Facebook**, publishing event and debate content for **500+ college students**.
- Coordinated with the **core team** on **event promotion** and regular audience communication, keeping campaigns consistent and on schedule.